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OpenLPTGUI: A user-friendly GUI for 3D Lagrangian particle tracking

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Summary

For complex flows, the flow velocity can be measured by capturing the motion of numerous small tracer particles that faithfully follow the flow (Nishino et al., 1989; Papantoniou & Dracos, 1989). Lagrangian particle tracking is one of the most established ways to extract the flow information, which offers access to long trajectories of tracer particles (Barta et al., 2024; Schanz et al., 2016; Shnapp, 2022; Tan et al., 2020). These long tracks in the Lagrangian framework make this method widely applicable for problems such as particle dispersion, mixing, and many transport-related quantities (Godbersen et al., 2021; Masuk, Qi, et al., 2021; Qi et al., 2022; Schröder & Schanz, 2023; Tan, Zhong, et al., 2023; Tan & Ni, 2022).

Although the 2D tracking algorithm has been widely available (Ouellette, n.d.), the 3D tracking of a high concentration of particles with large image overlap and thus large ambiguities, computed in an efficient way on high-performance computing clusters and also with a user-friendly graphic interface, has not been available. Accomplishing this task requires efficient parallelization, memory management, and state-of-the-art algorithm to utilize intensity and temporal information for tracking. First of all, the algorithm of iterative particle reconstruction is used to effectively segment and triangulate particles at high concentrations (Wieneke, 2012). The temporal information can be utilized by assuming the temporal coherence of established tracks in predicting the position in the next frame, an algorithm called Shake-The-Box method (Schanz et al., 2016; Tan & Ni, 2022).

The optimized code, known as `OpenLPT`, has been made available on Github (Tan et al., 2020). However, despite its robustness, the code was written in C++ without a user-friendly API or GUI, which poses challenges for the community to compile and use. In this work, we present a completely-new version of this code with user-friendly GUI and python-based API, dedicated to making OpenLPT more accessible to the scientific community.

Statement of need

`pyOpenLPT` is an OpenLPT-affiliated Python package for three dimensional Lagrangian particle tracking. The source code, written in C++, is designed to handle dense particle tracking ($\sim O(5 \times 10^4)$). Compared with the old version (Tan et al., 2020), several new features have been added: (1) the polynomial camera model is added to the new code; (2) have more capability of calibraiton; and (3) flexibility with number of cameras and reduced cameras.

The new code has also been restructured and organized to be more modular and extendable, making it easier to incorporate new features, such as complex object identification, tracking, and simultaneous tracking of multiple types of objects. The Python package, `pyOpenLPT`,

wraps the low-level code written in C++, keeping its high efficiency without losing flexibility or ease-of-use in the user-interface. The installation for this package can be finished in several lines and is compatible to both Windows and Linux systems.

The GUI for `pyOpenLPT`, called `OpenLPTGUI`, is designed to offer a user-friendly graphical interface, making it accessible for users of all backgrounds to apply Lagrangian particle tracking for flow measurement. The interface features a clear, step-by-step workflow, and comprehensive documentation is included in the code folder. The workflow includes: calibration points extraction, camera calibration, image pre-processing, tracking, and camera parameter optimization (see Figure 1). A sample results with 50% of tracks is shown in Figure 2.

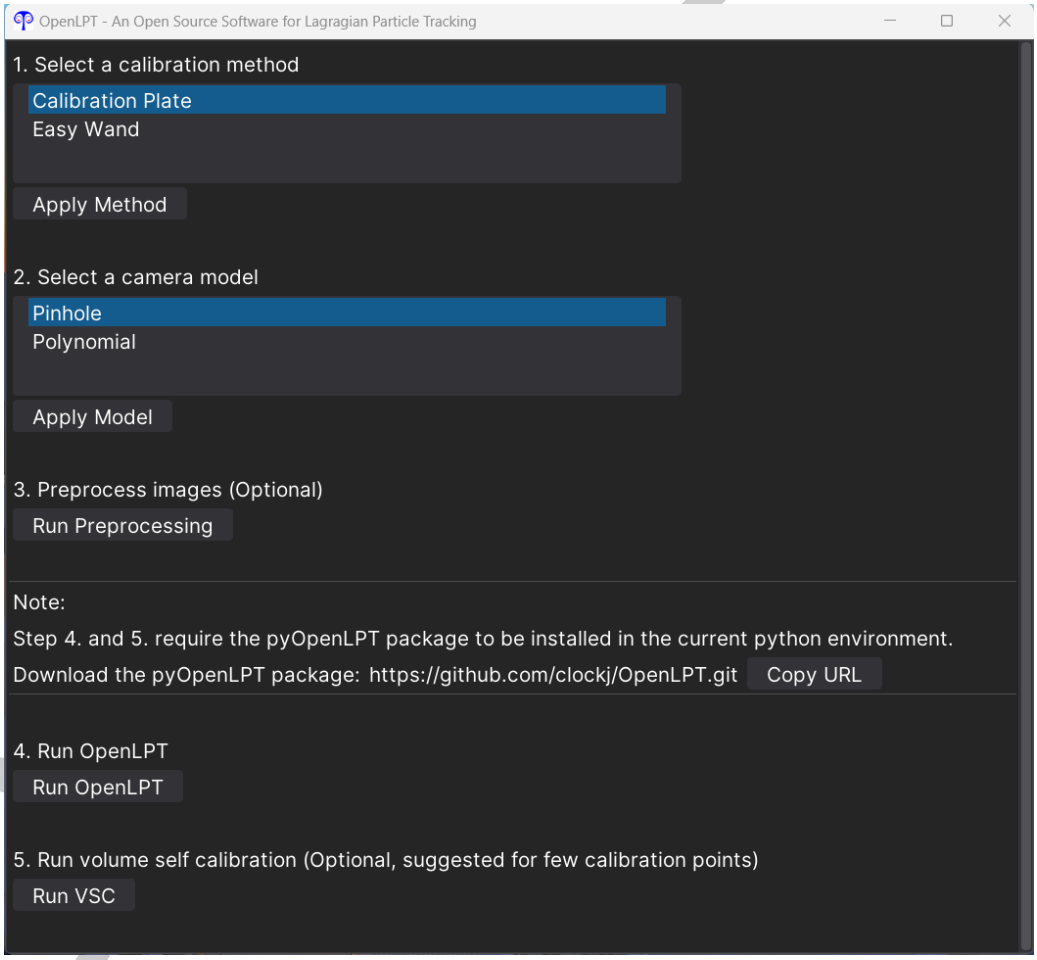

Figure 1: OpenLPTGUI main page.

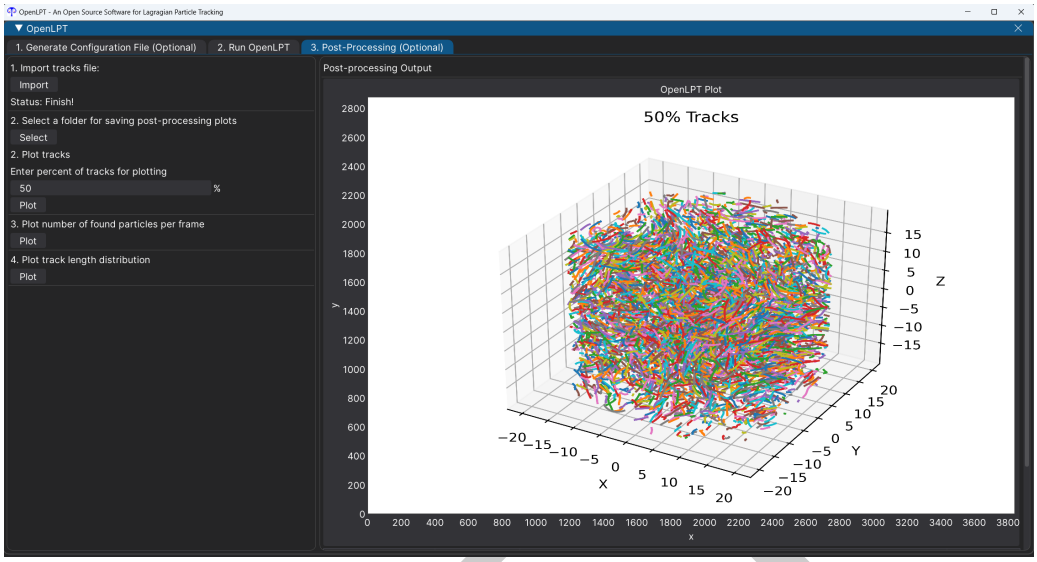

Figure 2: OpenLPTGUI post-processing for plotting 50% of tracks.

The code family (`OpenLPT`, `pyOpenLPT` and `OpenLPTGUI`) has already been used in a number of scientific publications (Masuk, Salibindla, et al., 2021a, 2021b; Masuk, Qi, et al., 2021; Qi et al., 2020, 2022; Salibindla et al., 2020, 2021; Tan, Xu, et al., 2023; Tan & Ni, 2022). With the new version of the code family, the advanced software can be more easily adopted by the experimental community, enabling more researchers to obtain high-quality 3D flow measurements and make exciting new discoveries.

Current capabilities

`pyOpenLPT`, currently in version 0.1.0, includes all the essential code required to extract 3D particle trajectories from image data. At present, the code does not support 2D tracking, nor does it integrate the functionality for tracking spherical objects (Tan, Zhong, et al., 2023), but both features are planned for inclusion in future versions.

Test dataset

A test dataset for tracking 3D tracer particles is provided at `test_STB`. Calibrated camera files are located in the subfolder `camFile`, and synthetic image files are located in the subfolder `imgFile`. Users need to modify the paths in the configuration files and image path files.

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